

ProServe X

Comprehensive Project Report

May 2026 | Version 1.1

Local Service Marketplace Platform

Django 6.0.5 | Python 3.13 | Bootstrap 5.3

1. Executive Summary

ProServe X is a full-stack Django web application that operates as a digital marketplace, connecting customers with verified local service providers across Pakistan. The platform supports 6 service categories - Carpentry, Electrical Work, Painting, Sweeping, Plumbing, and Gardening - and handles the complete lifecycle from provider discovery through booking, payment, and post-service review.

The system implements a dual-role authentication model (Customer / Provider) with granular access controls, role-specific dashboards, and a SQLite database comprising 7 interrelated models. The frontend delivers a responsive experience via Bootstrap 5.3 with custom dark-theme styling, glassmorphism UI components, and AOS animations. The project is structured into 5 modular Django apps totaling 12 view functions, 18 URL routes, and 13 HTML templates.

2. Technology Stack

2.1 Software Stack

Layer	Technology
Backend	Python 3.13, Django 6.0.5, WSGI
Database	SQLite 3 (dev) / PostgreSQL (production)
User Model	Custom AbstractUser with role/city/phone/profile_pic
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5.3, AOS 2.3
Form Rendering	Django ModelForms + django-crispy-forms (Bootstrap 4)
Media	Pillow 12.2, local filesystem storage
Static Assets	Custom CSS (592 lines), custom JS (12 lines)
Server	Django dev server on port 8080

2.2 Django Configuration

AUTH_USER_MODEL: accounts.CustomUser

INSTALLED_APPS: admin, auth, contenttypes, sessions, messages, staticfiles, crispy_forms, accounts, providers, bookings, reviews, core

MIDDLEWARE: Security, Session, Common, CsrfView, Authentication, Message, XFrameOptions

Template Engine: DjangoTemplates with request, auth, messages, + categories_processor

Password Validators: UserAttributeSimilarity, MinimumLength, CommonPassword, NumericPassword

Auth URLs: LOGIN_URL=login, LOGIN_REDIRECT_URL=dashboard, LOGOUT_REDIRECT_URL=home

Static: STATICFILES_DIRS=static/, STATIC_ROOT=staticfiles/

Media: MEDIA_URL=/media/, MEDIA_ROOT=media/

3. Architecture Overview

3.1 Application Modules

The platform is composed of 5 modular Django apps:

App	Responsibility
core	Homepage rendering, provider search/filter, global context processor
accounts	Custom user model, registration, login/logout, role-based dashboards
providers	ProviderProfile CRUD, skills, portfolio images, availability schedules
bookings	Booking lifecycle, status transitions, payment processing, transaction history
reviews	Review model, rating aggregation, review form logic (in providers.views)

3.2 Data Flow

Browser -> HTTP Request -> urls.py (URL Dispatcher) -> views.py (Business Logic) -> models.py (Database) -> Template -> HTML Response

3.3 Business Workflow

Register -> Login -> Browse/Search Providers -> View Profile -> Submit Booking
-> Provider Accepts -> Service Delivered -> Customer Pays -> Customer Reviews

3.4 Complete Route Map

Prefix	App	Routes
/	core	home (GET /), search (GET /search/)
/accounts/	accounts	register, login, logout, dashboard
/providers/	providers	list, by-category, detail, edit-profile
/bookings/	bookings	my-bookings, status-update, pay, payment-panel
/admin/	admin	Django admin console (all models)
/media/	dev	Media file serving (DEBUG=True only)

Total: 18 URL patterns across 5 apps + admin + media serving

3.5 Context Processors

core.context_processors.categories_processor injects SERVICE_CATEGORIES (6 categories) into all templates as {{ categories }}, powering dynamic navigation and footer links.

4. Data Model

7 Django models with the following entity relationships:

```
CustomUser (1) -----> (1) ProviderProfile (if role=provider)
|
|-----> (N) Skill
|
|-----> (N) WorkImage
|
|-----> (N) Availability
+-----> (N) Booking (as customer)
+-----> (N) Booking (as provider)
+-----> (N) Review (as customer)
+-----> (N) Review (as provider)
Booking (1) -----> (1) Review
```

4.1 accounts.CustomUser (extends AbstractUser)

Field	Type	Max	Details
id	AutoField	-	Primary Key
username	VARCHAR	150	Unique, required
password	VARCHAR	128	Hashed
email	EmailField	254	Required
role	CharField	20	customer / provider (default: customer)
phone	CharField	15	Blank allowed
city	CharField	100	200+ Pakistan cities (choices)
profile_pic	ImageField	-	Upload: profile_pics/, nullable
is_active	BooleanField	-	Default: True
date_joined	DateTimeField	-	Auto-set on creation

4.2 providers.ProviderProfile

Field	Type	Max	Details
id	AutoField	-	Primary Key
user	OneToOneField	-	FK -> CustomUser, CASCADE
service_category	CharField	100	6 choices (carpenter/electrician/painter/sweeper/plumber/gardener)
description	TextField	1000	Optional, blank=True
experience_years	PositiveIntegerField	-	Default: 0
price_per_hour	DecimalField	10,2	Default: 0.00
city	CharField	100	200+ Pakistan cities (choices)
is_available	BooleanField	-	Default: True
average_rating	DecimalField	3,2	Default: 0.00
created_at	DateTimeField	-	Auto-set on creation

4.3 Provider Supporting Models

Model	Fields	Details
Skill	provider (FK), name	Individual skills on a provider profile

WorkImage	provider (FK), image, caption	Portfolio images uploaded to work_samples/
Availability	provider (FK), day, start_time, end_t	Weekly schedule time slots

4.4 bookings.Booking

Field	Type	Max	Details
id	AutoField	-	Primary Key
customer	ForeignKey	-	FK -> CustomUser (bookings_as_customer)
provider	ForeignKey	-	FK -> CustomUser (bookings_as_provider)
service_date	DateField	-	Scheduled service date
service_time	TimeField	-	Scheduled service time
status	CharField	20	pending/accepted/completed/cancelled
address	TextField	-	Service location address
notes	TextField	-	Optional, blank=True
total_amount	DecimalField	10,2	Default: 0.00
payment_method	CharField	20	cash/card/online (default: cash)
payment_status	CharField	20	unpaid/paid/refunded (default: unpaid)
created_at	DateTimeField	-	auto_now_add=True
updated_at	DateTimeField	-	auto_now=True

4.5 reviews.Review

Field	Type	Max	Details
id	AutoField	-	Primary Key
booking	OneToOneField	-	FK -> Booking (related_name=review)
customer	ForeignKey	-	FK -> CustomUser (reviews_given)
provider	ForeignKey	-	FK -> CustomUser (reviews_received)
rating	PositiveSmallInt	-	1-5 (choices built from range(1,6))
comment	TextField	500	Optional, blank=True
created_at	DateTimeField	-	auto_now_add=True

5. Views & Business Logic

12 view functions across 5 apps:

5.1 core.views (2 views)

View	Route	Logic
home_view	GET /	Extracts 6 category slugs; queries top-6 available providers by -average_rating; renders index.html
search_view	GET /search/	Filters by q (icontains on username), category, city; passes PAKISTAN_CITIES to template for dropdown

5.2 accounts.views (4 views)

View	Route	Logic
register_view	GET/POST /register/	POST: validates form, saves user, auto-login; if role=provider auto-creates ProviderProfile. GET: empty form
login_view	GET/POST /login/	POST: authenticate -> login -> redirect dashboard; failure -> messages.error
logout_view	GET /logout/	logout(request) -> redirect(home)
dashboard_view	GET /dashboard/	@login_required. Provider: bookings as provider + reviews. Customer: bookings as customer

5.3 providers.views (3 views)

View	Route	Logic
provider_list	GET / or /category/	Filters is_available=True; optional category param; passes category slugs to template
provider_detail	GET/POST /<pk>/	GET: profile + booking form + review form. POST book: creates Booking; POST review: creates Review, recalcu
edit_profile	GET/POST /edit/	@login_required. Gets-or-creates ProviderProfile; validates ProviderProfileForm; saves

5.4 bookings.views (4 views)

View	Route	Logic
my_bookings	GET /	@login_required. Role-aware booking list ordered by -created_at
update_booking_status	GET /update/<pk>/<status>/	@login_required. Verifies ownership (provider); accepts accepted/completed/cancelled
make_payment	GET/POST /pay/	@login_required. Verifies ownership (customer); prevents double-payment; method + status update
payment_panel	GET /payments/	@login_required. Role-aware; computes total_earned + total_pending; supports ?payment_status= filter

6. Forms & Validation

6.1 UserRegisterForm (accounts.forms)

Extends UserCreationForm. Fields: username, email, phone, city (ChoiceField from PAKISTAN_CITIES), role (RadioSelect), service_category (conditional, Select, required=False), password1, password2.

- role uses RadioSelect widget with class role-radio for styled role cards
- service_category uses Select with class profession-select, shown conditionally via JS
- city includes empty placeholder option for user experience

6.2 ProviderProfileForm (providers.forms)

ModelForm for ProviderProfile. Fields: service_category, description (Textarea rows=4), experience_years, price_per_hour, city (ChoiceField from PAKISTAN_CITIES with placeholder), is_available.

6.3 BookingForm (bookings.forms)

ModelForm for Booking. Fields: service_date (DateInput type=date), service_time (TimeInput type=time), address (Textarea rows=3), notes (Textarea rows=3), payment_method (Select with class form-select).

6.4 ReviewForm (reviews.forms)

ModelForm for Review. Fields: rating (RadioSelect choices 1-5), comment (Textarea rows=4).

7. Feature Set

7.1 Authentication & Role Management

Dual-role registration via radio button selection with conditional field display. Provider accounts auto-create a ProviderProfile upon registration. Protected views use @login_required with custom LOGIN_URL, LOGIN_REDIRECT_URL, and LOGOUT_REDIRECT_URL settings.

7.2 Provider Ecosystem

Providers maintain rich profiles including:

- Service category (6 options) and description (max 1000 chars)
- Skills list (one-to-many via Skill model)
- Portfolio images with captions (WorkImage, stored in work_samples/)
- Weekly availability schedule (day, start_time, end_time)
- Experience years, hourly rate, city, and availability toggle
- Average rating auto-calculated from all customer reviews

7.3 Booking Lifecycle

Pending -> Accepted -> Completed -> Reviewable

-> Cancelled (by provider)

Customers initiate bookings with date, time, address, notes, and payment method. Providers accept, complete, or cancel. Payments tracked as cash/card/online with unpaid/paid/refunded status.

7.4 Search & Discovery

Multi-parameter search: provider name (icontains), service category (exact), city (exact). Homepage shows top-6 providers by average_rating. Category pills for quick filtering.

7.5 Ratings & Feedback

Post-service reviews require a completed booking. Ratings 1-5 stars with optional comment. Average rating recalculated as mean of all ratings after each submission.

7.6 Dashboards

Role	Content
Customer	Booking history table; stats (total, completed, pending, cancelled)
Provider	Incoming requests with accept/complete/cancel; earnings summary; reviews

7.7 Payment Panel

Aggregated view showing total_earned (paid) and total_pending_amount (unpaid). Supports filtering by ?payment_status=paid|unpaid. Displays paid_count and unpaid_count statistics.

8. User Interface & Templates

13 templates extending base.html, using Bootstrap 5.3, Bootstrap Icons, and AOS 2.3:

Template	Description
base.html	Master: dark navbar (role-aware menu), auto-fade alerts, footer with categories
index.html	Hero section, 6 category cards (glassmorphism), top-providers grid, about, CTA
login.html	Centered auth card with username/password fields, link to registration
register.html	Role selection cards (Customer/Provider), conditional profession field
dashboard.html	Customer stats + booking history table
provider_dashboard.html	Provider stats, booking table with actions, reviews sidebar
provider_list.html	Category pills, provider cards (avatar, name, category, rating, price)
provider_detail.html	Profile card, skills, gallery, avail table, booking/review forms
my_bookings.html	Role-aware booking table with status badges, action buttons
payment.html	Payment form with amount + method selector
payment_panel.html	Stats cards + filterable transaction history table
edit_profile.html	Provider profile edit form with all fields
search_results.html	Search filters + results grid with provider cards

8.1 Visual Identity

- Dark gradient theme with navy/indigo hero section using CSS custom properties
- Glassmorphism effects: backdrop-filter blur, semi-transparent backgrounds on cards
- Hover animations: scale transforms on cards, color transitions, glow effects
- Responsive: mobile hamburger menu, fluid grids, stacked layouts on small screens
- Bootstrap Icons for UI elements: tools, user, calendar, credit-card, star ratings
- AOS scroll animations for fade/slide effects on page sections
- Custom scrollbar via webkit pseudo-elements
- Cache-busting (?v=5) on style.css for development

8.2 base.html Structure

- Navbar: brand (bi-tools + ProServe X), links (Home, Providers, Search), role-aware dashboard, dropdown user menu (dashboard, edit profile if provider, logout), login/signup for guests
- Messages: Django messages rendered as Bootstrap alerts with dismiss button
- Footer: brand description, quick links, 6 dynamic category links, contact info, copyright
- Scripts: Bootstrap 5.3 JS, AOS JS, custom main.js (AOS init + tooltip bootstrap)

9. Deployment Guide

9.1 Development Setup

```
python -m venv venv
venv\Scripts\activate
pip install django crispy-bootstrap4 pillow
python manage.py migrate
python manage.py runserver 8080
```

Convenience scripts: start_server.bat (opens browser + runs server) and start_server.vbs (silent launch).

9.2 Production Checklist

- Set DEBUG = False and secure SECRET_KEY via environment variable
- Configure ALLOWED_HOSTS with production domain(s)
- Replace SQLite with PostgreSQL using dj-database-url
- Add whitenoise to MIDDLEWARE for static file serving
- Offload media files to cloud storage (S3, Cloudinary)
- Enforce HTTPS with security middleware settings
- Run python manage.py migrate && python manage.py collectstatic
- Create superuser for Django admin access

9.3 Testing

```
python manage.py test accounts providers bookings reviews core
```

10. Project Metrics

Metric	Value
Django applications	5 (core, accounts, providers, bookings, reviews)
Database models	7 (CustomUser, ProviderProfile, Skill, WorkImage, Availability, Booking, Review)
URL endpoints	18 (5 core + 4 accounts + 4 providers + 4 bookings + 1 admin)
View functions	12 (2 core + 4 accounts + 3 providers + 4 bookings)
HTML templates	13 (base + 12 child templates)
Form classes	4 (UserRegister, ProviderProfile, Booking, Review)
Service categories	6 (carpenter, electrician, painter, sweeper, plumber, gardener)
Supported cities	200+ across Punjab, Sindh, KP, Balochistan, Islamabad, GB, AJK
Custom CSS	592 lines of custom styling
Custom JavaScript	12 lines (AOS init + tooltip bootstrap)
Python version	3.13
Django version	6.0.5
Database	SQLite 3 (dev), 1 file: db.sqlite3

Launch scripts	2 (start_server.bat, start_server.vbs)
Contact email	smadadali4@gmail.com

11. Conclusion

ProServe X is a production-ready service marketplace platform with a clean, modular architecture, comprehensive role-based access control, and a complete booking-to-payment-to-review workflow. The 5-app separation ensures maintainability and clear responsibility boundaries.

The frontend delivers a polished, responsive dark-themed experience with Bootstrap 5, AOS animations, and glassmorphism design patterns. The backend provides robust data modeling with 7 interrelated models, comprehensive form validation via 4 form classes, and role-aware business logic across 12 view functions and 18 URL endpoints.

The platform is deployment-ready with minor production configuration changes including database upgrade, static file serving via Whitenoise, and HTTPS enforcement. The modular architecture supports future enhancements such as real-time notifications, admin analytics dashboards, mobile REST API endpoints, and multi-language support.

End of Report | ProServe X | May 2026 | Version 1.1